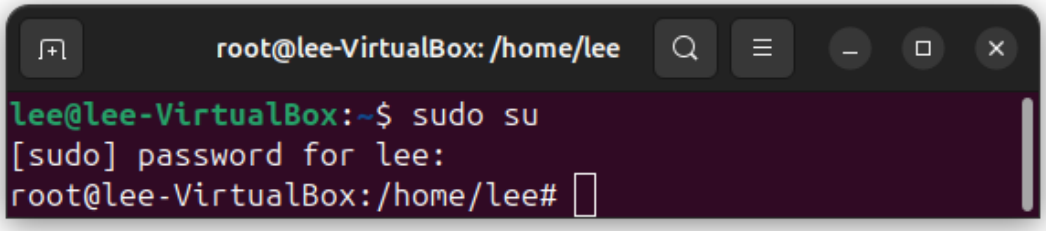
Lab 3: Authentication, Authorization, Access Control

1. Adding Groups, Users and Passwords

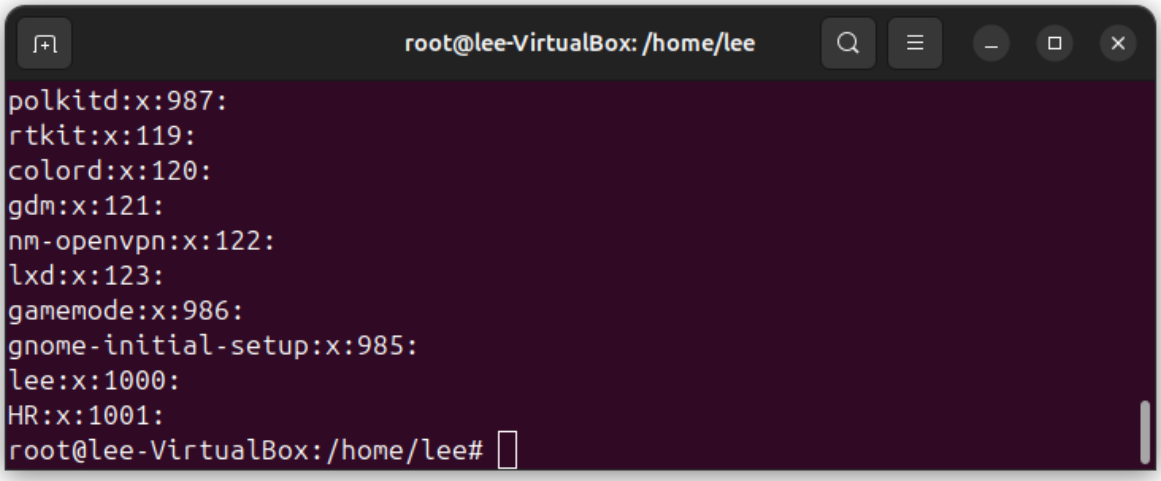
1) Open a new Terminal window by clicking on the Terminal icon located on the left menu pane.

2) Escalate privileges to the *root* level by typing the command below followed by pressing **Enter**.
   `sudo su`



3) Add a new group named **HR**.
   `groupadd HR`

4) Verify the new group has been added to the group file list.
   `cat /etc/group`
   The new group HR will be added to the bottom of the /etc/group file with a group ID of 1001.



5) Add a new user named **jenny**.
   `adduser jenny`

```
root@lee-VirtualBox: /home/lee
root@lee-VirtualBox:/home/lee# adduser jenny
info: Adding user `jenny' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jenny' (1002) ...
info: Adding new user `jenny' (1002) with group `jenny (1002)' ...
info: Creating home directory `/home/jenny' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for jenny
Enter the new value, or press ENTER for the default
        Full Name []: jenny
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `jenny' to supplemental / extra groups `users' ...
info: Adding user `jenny' to group `users' ...
root@lee-VirtualBox:/home/lee#
```

6) Place the user **jenny** in the newly created **HR** group.
   `usermod -G HR jenny`

7) Add another new user named **joe**.
   `adduser joe`

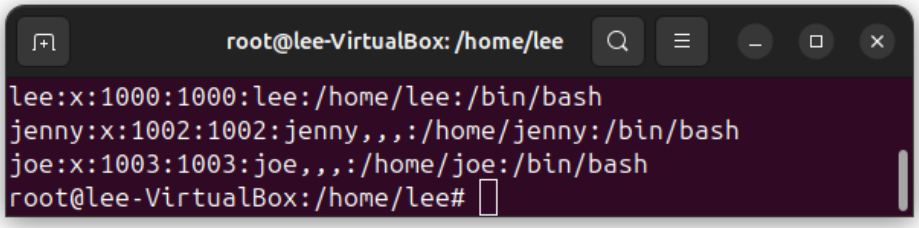
```
root@lee-VirtualBox: /home/lee
root@lee-VirtualBox:/home/lee# adduser joe
info: Adding user `joe' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `joe' (1003) ...
info: Adding new user `joe' (1003) with group `joe (1003)' ...
info: Creating home directory `/home/joe' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for joe
Enter the new value, or press ENTER for the default
        Full Name []: joe
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `joe' to supplemental / extra groups `users' ...
info: Adding user `joe' to group `users' ...
root@lee-VirtualBox:/home/lee#
```

8) Place the user **joe** in the **HR** group.

```
usermod -G HR joe
```

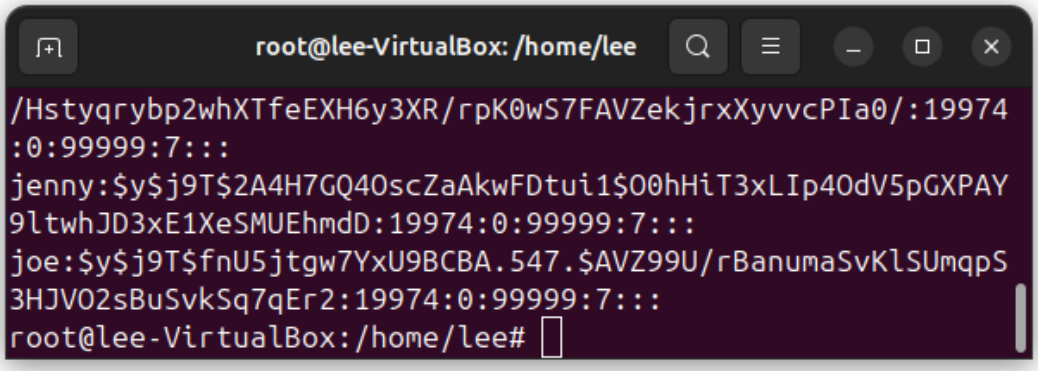
9) Verify the newly created users in the **passwd** file.

```
cat /etc/passwd
```



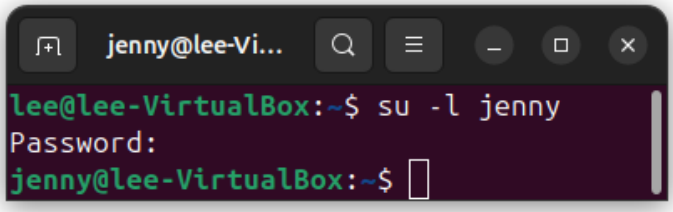
10) View the created users in the **shadow** file.

```
cat /etc/shadow
```

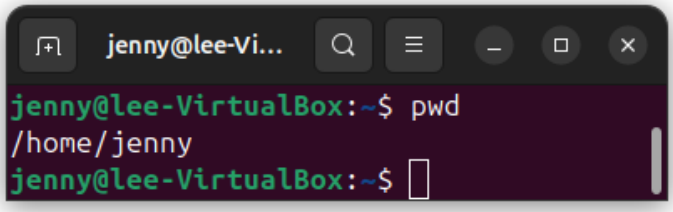


2. Symbolic Permissions

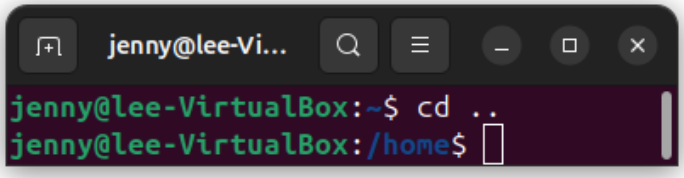
1) Open another Terminal and login the screen as a user jenny.
   `su -l jenny`



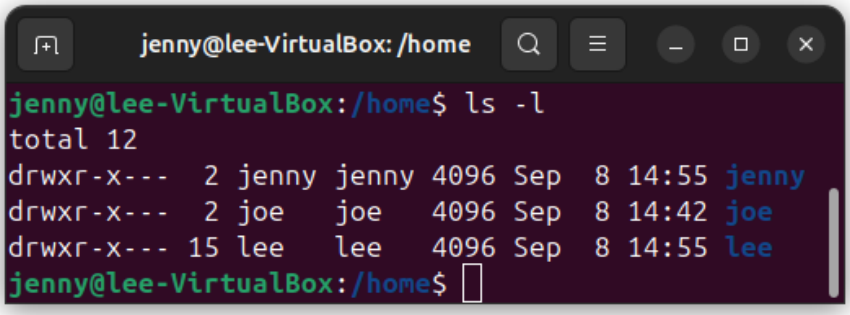
2) View your present directory.
   `pwd`



3) Go back one directory level to the **/home** directory.
   `cd ..`



4) List all directories and their permissions.
   `ls -l`



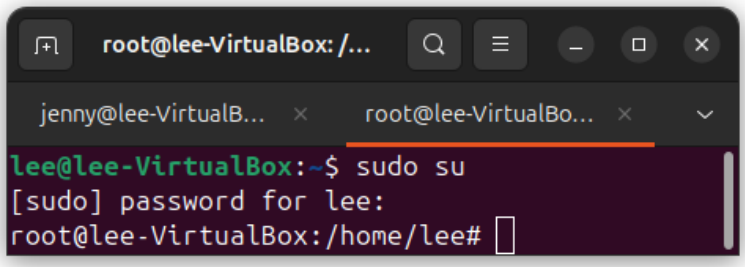
The Linux operating system has a total of 10 letters or dashes in the permissions fields:

- The first field is a dash for a file and a d for a directory (folder)
- The 2nd through 4th fields are for the user
- The 5th through 7th fields are for the group
- The 8th through 10th fields are for others (accounts other than those in the group)

5) Enter *Joe's folder* as Jenny by typing the command below.
```
cd joe
```

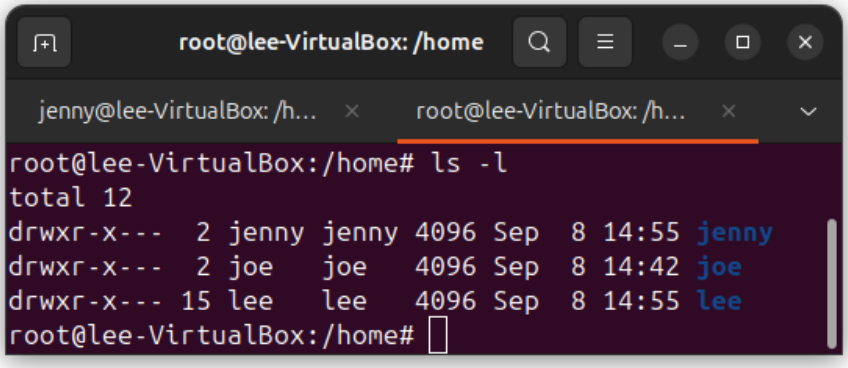
Notice that we are not able to go into *Joe's home folder*.

6) Open a new Terminal window and escalate privileges to the *root* level by typing the command below followed by pressing **Enter**.
```
sudo su
```



7) Change to the /home directory.
```
cd /home
```

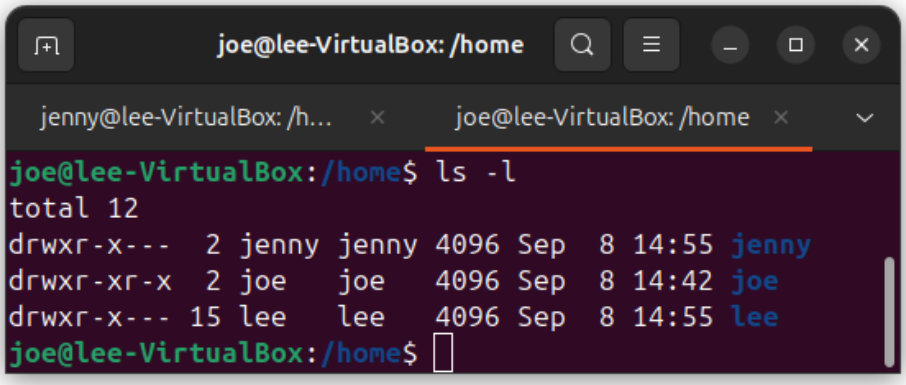
8) List all current directories with their permissions.
```
ls -l
```



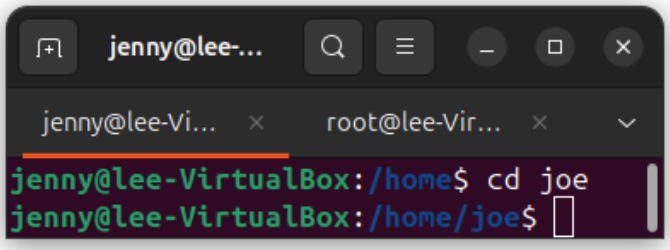
Take note of the "*other*" field for Joe's folder, notice that it is currently set with a symbol (-), which doesn't make it available to execute for "*other*" users.

9) Change the "other" permission on joe's folder by making it *non-executable*.
```
chmod o+rx joe
```

10) List the directories once more with their respective permissions.



11) Now that we switched to jenny's *Terminal* session, attempt to go into joe's folder once more.
    `cd joe`



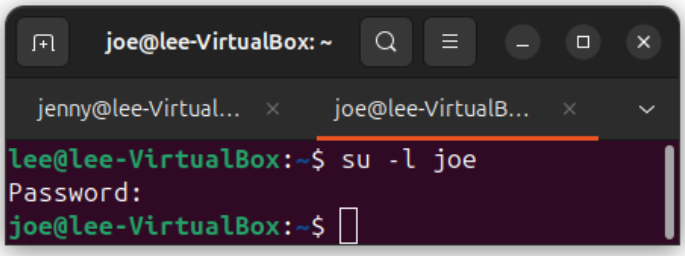
Notice that we do have permission to do so.

The chart below shows examples of other ways the *chmod* command can be used:

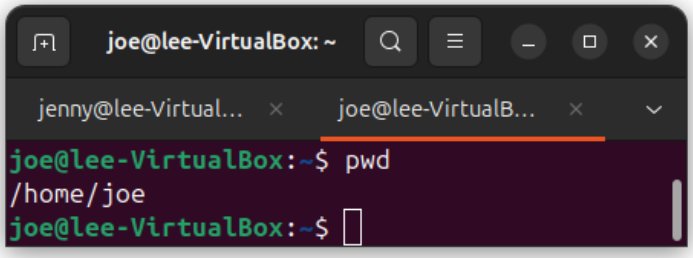
chmod command	Results
chmod u+rwx	Adds read, write and execute permissions for the user
chmod u+rw	Adds read and write permission for the user
chmod o+r	Adds read permission for others
chmod g-rwx	Removes read, write and execute permissions for the group

3. Absolute Permission

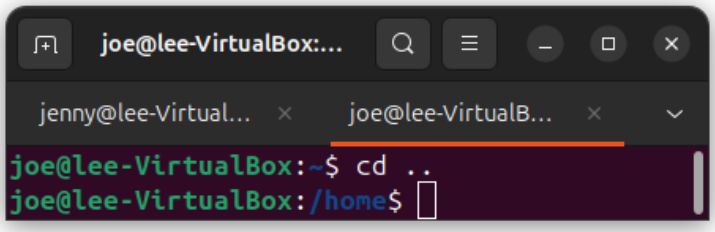
1) Open another Terminal and login the screen as a user **joe** with the password
   `su -l joe`



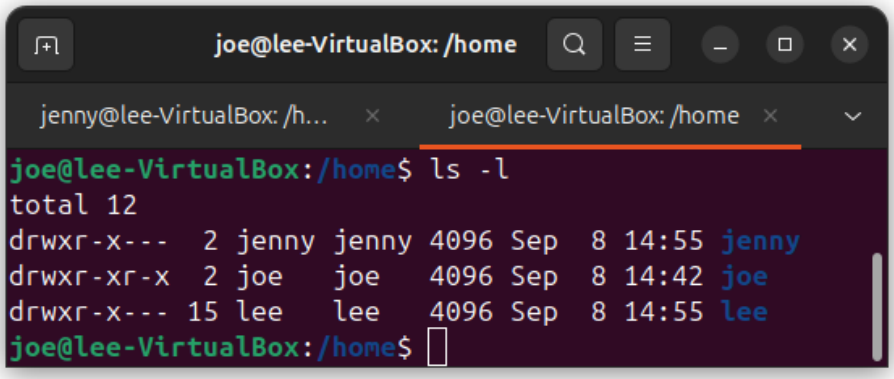
2) Print your current working directory.
   `pwd`



3) Go back one directory level to the */home* directory.
   `cd ..`



4) List all directories and their permissions in the current working directory.
   `ls -l`



The other way of assigning permissions besides using symbolic permissions is the use of absolute permissions. Absolute permissions use a three-digit octal number to represent the permissions for owner, group and others.

The table below outlines each absolute value and its corresponding permissions:

Number	Permissions
7	Read, Write and Execute
6	Read and Write
5	Read and Execute
4	Read
3	Write and Execute
2	Write
1	Execute
0	None

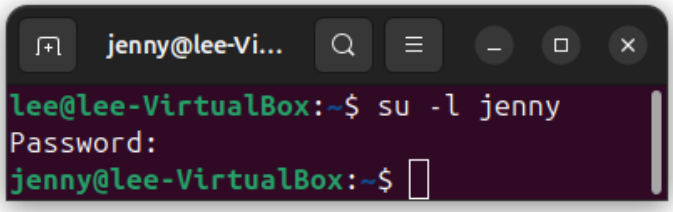
By typing the command, chmod 764 <examplefile>, the examplefile will be assigned the follow permissions:

- The user will get Read, Write and Execute permissions
- The group will get Read and Write permissions
- Others will get Read Access

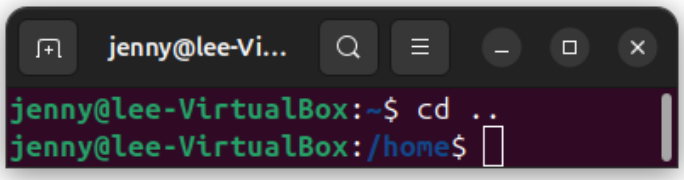
Breakdown of how 764 represents these permissions:

Digit	Binary Equivalent	Permission
7 (user)	111	1-Read 1-Write 1-Execute
6 (group)	110	1-Read 1-Write 0-No Execute
4 (others)	100	1-Read 0-No Write 0-No Execute

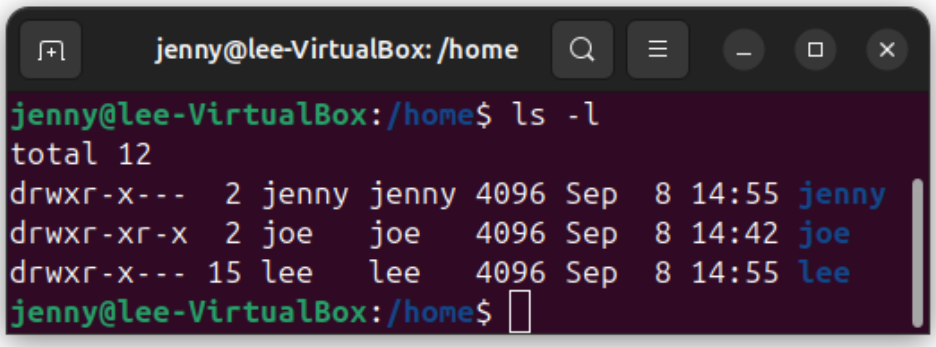
5) Open another Terminal and login the screen as a user jenny.
   `su -l jenny`



6) Go back one directory level to the **/home** directory.
   `cd ..`

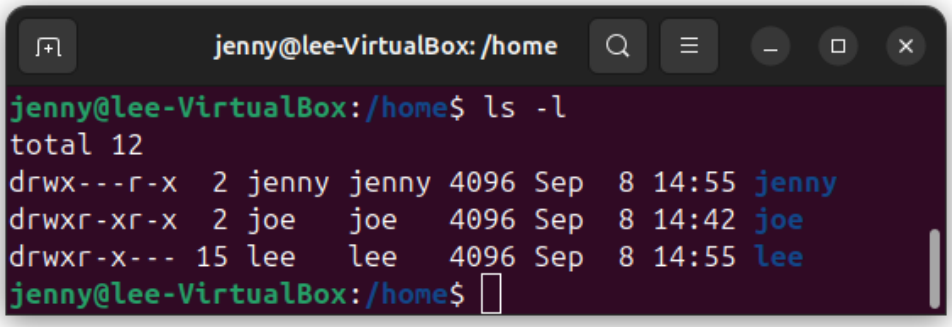


7) List all directories and their permissions.
   `ls -l`

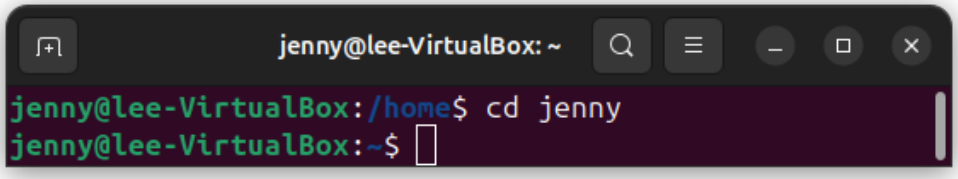


8) Modify the "*others*" field for jenny's folder so that others will be able read and execute but not write while still maintaining the "user" field to read, write, and execute.
   `chmod 705 jenny`

9) List the file permissions of the current directory to see that the absolute changes were made.
   `ls -l`



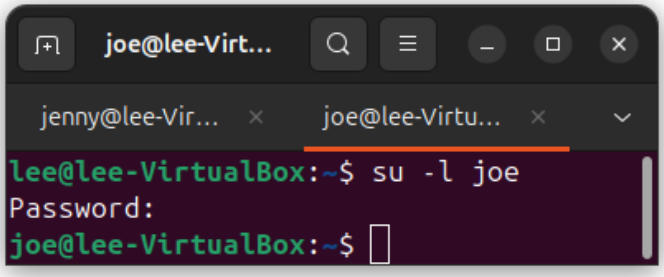
10) Change to the */home/jenny* directory.
   `cd jenny`



11) Create a simple text file named test.txt using touch.

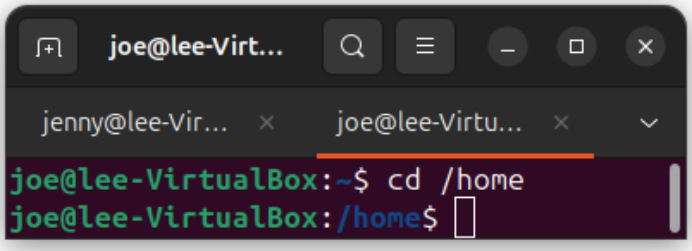
```
touch test.txt
```

12) login the screen as a user joe.



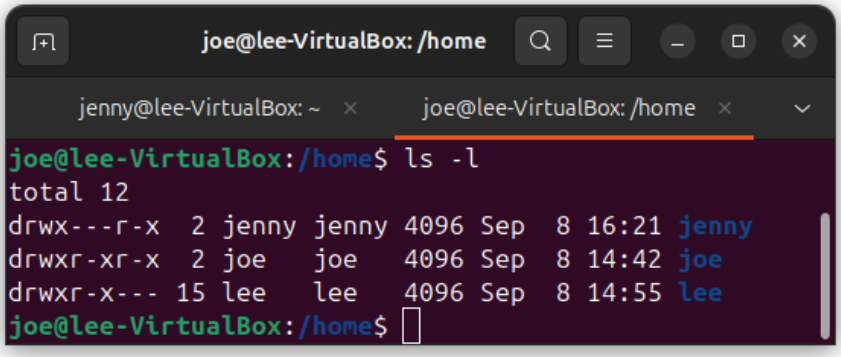
13) Change to the **/home** directory.

```
cd /home
```



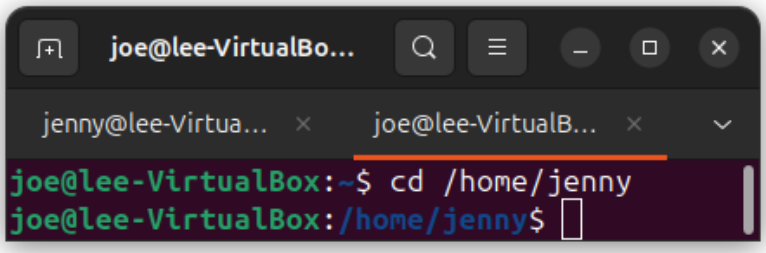
14) List all directories with their respective permissions.

```
ls -l
```

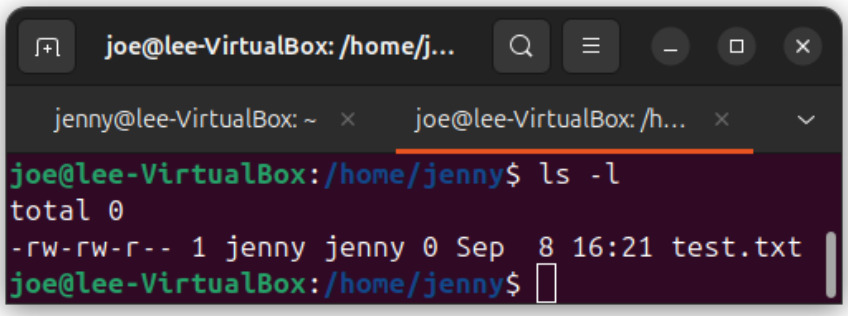


15) Change to the **/home/jenny** directory.

```
cd /home/jenny
```

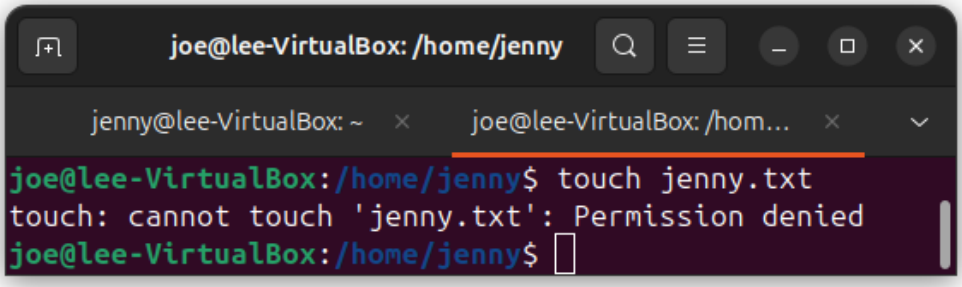


16) List all files in the current directory.
```
ls -l
```



Notice that we are able to enter Jenny's folder and read the files within the directory. We are able to see the *test.txt* file.

17) Attempt to create a file.
```
touch jenny.txt
```



Notice we do not have permission to create the file.

18) **Close** all remaining windows.

Provide answers to these questions (in this document. Please use a different color for your responses)

1. **What does # indicate in the command prompt in Linux? # (pound) means that the current user has root permissions.**
2. **Fill in the last column in the table with the absolute permissions (octal code) for the directory CYBR2106 and files assignment.txt and exam.txt.**

	owner	group	other	Octal code
IT203	r-x	rwx	r--	**574**
assignment.txt	rwx	rwx	r-x	**775**
exam.txt	r-x	rwx	---	**570**

Submit your assignment though Canvas.

Grading Rubric:

	MaxPoints
Screenshots from Lab	4
Answers to questions	4
Spelling/Grammar	2
Total	10

```
HR:x:1001:
root@desktop-VMware-Virtual-Platform:/home/desktop# adduser jenny
info: Adding user `jenny' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jenny' (1002) ...
info: Adding new user `jenny' (1002) with group `jenny (1002)' ...
info: Creating home directory `/home/jenny' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for jenny
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `jenny' to supplemental / extra groups `users' ...
info: Adding user `jenny' to group `users' ...
root@desktop-VMware-Virtual-Platform:/home/desktop# usermod -G HR jenny
root@desktop-VMware-Virtual-Platform:/home/desktop# adduser joe
info: Adding user `joe' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `joe' (1003) ...
info: Adding new user `joe' (1003) with group `joe (1003)' ...
info: Creating home directory `/home/joe' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for joe
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
```

```
root@desktop-VMware-Virtual-Platform:/home/desktop# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
```

```
desktop:$6$VPC5tkw40UPxMPxg$qp6ahANpzWktKbBeYL4EOHo3PYrWD82XOcKMaHwCyLc8GhiTs/Owp6RVN6ol8lU4iFa1HVvxQAm.atNUFyP2j0:20347:0:99999:7:::
jenny:$y$j9T$PC/Aj2o.383EXxy1.vmlA.$VliGZOK5RDZGjN5h8Q8B2zhVQWXzF6S3NACFuM5dhS5:20347:0:99999:7:::
joe:$y$j9T$VV.JBRGPWboyq.MJrifEM/$Bc1z06ZavuW0kBCr5bL2qWvE2WvclGorebUEr8sweGD:20347:0:99999:7:::
```

```
jenny@desktop-VMware-Virtual-Platform: ~    joe@desktop-VMware-Virtual-Platform: /hom...
desktop@desktop-VMware-Virtual-Platform:~$ su -l jenny
Password:
jenny@desktop-VMware-Virtual-Platform:~$ cd ..
jenny@desktop-VMware-Virtual-Platform:/home$ ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwxr-x---  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-xr-x  2 joe     joe     4096 Sep 16 15:29 joe
jenny@desktop-VMware-Virtual-Platform:/home$ chmod 705 jenny
jenny@desktop-VMware-Virtual-Platform:/home$ ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwx---r-x  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-xr-x  2 joe     joe     4096 Sep 16 15:29 joe
jenny@desktop-VMware-Virtual-Platform:/home$ cd jenny
jenny@desktop-VMware-Virtual-Platform:~$ touch test.txt
jenny@desktop-VMware-Virtual-Platform:~$
```

```
jenny@desktop-VMware-Virtual-Platform: ~    joe@desktop-VMware-Virtual-Platform: /hom...
desktop@desktop-VMware-Virtual-Platform:~$ su -l joe
Password:
joe@desktop-VMware-Virtual-Platform:~$ cd /home
joe@desktop-VMware-Virtual-Platform:/home$ ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwx---r-x  2 jenny   jenny   4096 Sep 16 15:41 jenny
drwxr-xr-x  2 joe     joe     4096 Sep 16 15:29 joe
joe@desktop-VMware-Virtual-Platform:/home$ cd /home/jenny
joe@desktop-VMware-Virtual-Platform:/home/jenny$ ls -l
total 0
-rw-rw-r-- 1 jenny jenny 0 Sep 16 15:41 test.txt
joe@desktop-VMware-Virtual-Platform:/home/jenny$ touch jenny.txt
touch: cannot touch 'jenny.txt': Permission denied
joe@desktop-VMware-Virtual-Platform:/home/jenny$
```

```
jenny@desktop-VMware-Virtual-Platform: /home/joe
desktop@desktop-VMware-Virtual-Platform:~$ su -l jenny
Password:
jenny@desktop-VMware-Virtual-Platform:~$ cd ..
jenny@desktop-VMware-Virtual-Platform:/home$ ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwxr-x---  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-x---  2 joe     joe     4096 Sep 16 15:29 joe
jenny@desktop-VMware-Virtual-Platform:/home$ cd joe
-bash: cd: joe: Permission denied
jenny@desktop-VMware-Virtual-Platform:/home$ cd joe
jenny@desktop-VMware-Virtual-Platform:/home/joe$
```

```
desktop@desktop-VMware-Virtual-Platform:~$ sudo su
[sudo] password for desktop:
root@desktop-VMware-Virtual-Platform:/home/desktop# cd /home
root@desktop-VMware-Virtual-Platform:/home# ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwxr-x---  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-x---  2 joe     joe     4096 Sep 16 15:29 joe
root@desktop-VMware-Virtual-Platform:/home# chmod o+rx joe
root@desktop-VMware-Virtual-Platform:/home# ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwxr-x---  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-xr-x  2 joe     joe     4096 Sep 16 15:29 joe
root@desktop-VMware-Virtual-Platform:/home#
```

```
desktop@desktop-VMware-Virtual-Platform:~$ su -l joe
Password:
joe@desktop-VMware-Virtual-Platform:~$ pwd
/home/joe
joe@desktop-VMware-Virtual-Platform:~$ cd ..
joe@desktop-VMware-Virtual-Platform:/home$ ls -l
total 12
drwxr-x--- 16 desktop desktop 4096 Sep 16 15:06 desktop
drwxr-x---  2 jenny   jenny   4096 Sep 16 15:28 jenny
drwxr-xr-x  2 joe     joe     4096 Sep 16 15:29 joe
joe@desktop-VMware-Virtual-Platform:/home$
```